Modelling Formula 1 Tyre Degradation and Optimising Pit-Stop Strategy

From-scratch numerical methods, validated degradation models and a verified race-time optimiser for the Hungarian Grand Prix, 2022–2025

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GitHub repository: https://github.com/macuzzzz/F1-tire-Degradation-Model- (all numbers below come from running the code in this repository; the decision log DECISIONS.md is cited as D1–D92 and the challenge log CHALLENGES.md as C1–C24)

1. Engineering problem and objectives

A racing tyre loses grip as it wears, so lap time rises with tyre age. A pit stop restores the pace but costs about 21 s in the pit lane, so race strategy is a trade-off between running slower on worn tyres and paying for extra stops. A useful strategy model therefore needs a quantitative degradation curve for each tyre compound. Raw lap times cannot provide it directly, because degradation is mixed with other effects: the car becomes lighter as fuel burns (faster laps), the track rubbers in, safety cars and yellow flags slow laps, in- and out-laps include pit-lane time, and every car and driver has its own base pace.

Objective. From public race timing data: (i) remove the confounding effects (lap filtering, fuel correction, pit laps); (ii) fit a degradation model d(tyre age) per Pirelli compound; (iii) validate the models on laps and a race they were not fitted to; (iv) use the validated model in a race-time model to find the pit lap(s) that minimise total race time; and (v) test how sensitive that answer is to the uncertain inputs. All numerical methods (Euler, Heun, RK4, adaptive RK45, bisection, secant, Newton, finite differences, Gaussian elimination, least squares, KNN) are written from scratch with NumPy; SciPy and scikit-learn are used only as benchmarks.

Intended use and scope. The result is an engineering screening tool and a numerical-methods study, not an operational team strategy system. The scope is one circuit (Hungaroring), Pirelli compounds C3 and C4, seasons 2022–2025 and all teams. Car performance is not modelled: a separate base lap time per stint absorbs car, driver and compound pace (D39). Only free data are used, because F1's timing archive has required an F1 TV login since about 15 October 2026 (D14).

2. Inputs, outputs and data

Table 1. *Top-level inputs (all in config.yaml).*

Input	Unit / format	Valid range (checked by config.py)	Value used
Races: season, event, train/test role, source	YAML list	2018 to current year; OpenF1 ≥ 2023; ≥ 1 train and 1 test race; no duplicates	Hungary 2022–24 train, 2025 test
Pirelli compounds	C0–C6	known codes only	C3, C4
Fuel coefficient k_fuel	s per lap of fuel	≥ 0 and within [k_min, k_max]	0.046 (range 0.015–0.070)
Pit-loss override	s	> 0, or null = estimate from data	null → 20.89 s (median)
Outlier threshold k / min. stint	robust SD / laps	> 0 / integer ≥ 1	3.0 / 10 laps
Numerical and model settings, seed	see config.yaml	validated per section	seed 42

Table 2. *Top-level outputs.*

Output	Unit	Meaning and use	File (reports/)
Degradation model per compound + bootstrap SE	s, s/lap	lap-time increase vs tyre age; input to the strategy	phase6_inputs.csv
Validation metrics (RMSE, MAE, bias, R^2)	s, s/lap	accuracy on withheld races	phase5_cv_*.csv, phase5_test_results.csv
Optimal pit lap(s), total race time	lap, s	strategy ranking for a 70-lap race	phase6_strategy_ranking.csv
Sensitivity coefficients, break-even values	s, laps, –	which inputs can change the decision	phase7_*.csv
Audit, manifest, 24 figures	–	leakage/constraint checks, reproducibility hashes	audit_*.csv, figures/

Invalid input handling. The configuration is validated when it is loaded, with clear errors for wrong types, out-of-range values or missing keys. Every public function documents and checks its inputs and raises TypeError or ValueError (a review script confirms 213 public objects with complete NumPy docstrings, D86). Missing lap times are kept as NaN and removed by a cleaning rule; an unknown race raises RaceNotFoundError, and a blocked download raises RaceDataUnavailableError.

2.1 Data and preparation

Lap data come from FastF1 (2022, its free mirror) and the OpenF1 API (2023–2025), merged into one table and cached on disk; the raw data are never committed. Because SOFT/MEDIUM/HARD are relative labels, every stint is mapped to its Pirelli compound with a sourced allocation table (C3 was the MEDIUM in 2022 and the HARD in 2023–2025). Green-flag laps are detected from FastF1's track status (2022) or rebuilt from race-control messages (OpenF1). Eight cleaning rules are then applied in a fixed order, and each lap is counted against the first rule that removes it (Table 3).

Table 3. *Cleaning rules and the laps they removed.*

Rule (in order)	Why the lap cannot be used	Laps removed (4 races)
1 Missing lap time	no time recorded	7
2 First lap	standing start, pack racing	79
3 In- and out-laps	contain pit-lane time (used for the pit loss instead)	294
4 Not green flag	yellow flag, safety car, VSC or red flag during the lap	184
5 Inaccurate (FastF1 flag)	timing marked unreliable (none in these races)	0
6 Missing stint data	stint, compound or tyre age unknown (none)	0
7 Outlier within its stint	residual > 3 × 1.4826·MAD from a Theil–Sen line (traffic, mistakes)	313
8 Stint shorter than 10 laps	too few laps to separate offset and slope	114
Total	5,363 laps loaded → 4,372 clean laps (2022: 1012, 2023: 1013, 2024: 1118, 2025: 1229)	991

The outlier rule is robust on purpose: the Theil–Sen slope is the median of all pairwise slopes in a stint, and 1.4826·MAD estimates the standard deviation from the median absolute deviation, so a few slow laps cannot distort the line that judges them. **Fuel correction:** corrected time = t + k_fuel (n − 1) with k_fuel = 0.046 s per lap of fuel, i.e. every lap is expressed at the lap-1 fuel load. The value agrees with a joint fit on the training data (0.0456 s/lap) and with a rule of thumb (0.03 s/kg × 110 kg / 70 laps = 0.047); because track evolution also makes laps faster linearly, the coefficient contains both effects. **Pit loss:** in-lap plus out-lap minus two typical clean laps of the same driver, median of 105 green-flag stops in 2023–2025: 20.89 s (IQR 19.82–22.31 s).

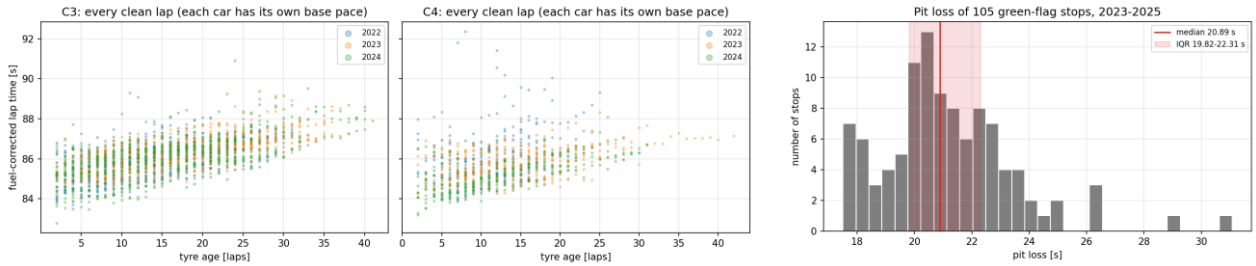


Figure 1. *Left: every clean training lap, fuel-corrected lap time against tyre age (each car has its own level, which the per-stint offsets remove). Right: the measured pit losses, 2023–2025, with median and IQR.*

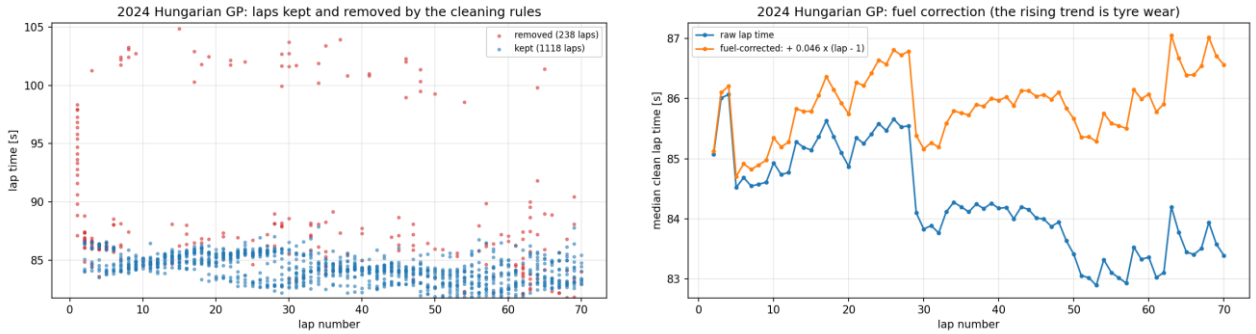


Figure 2. *2024 race: laps kept and removed by the cleaning rules (left); median lap time per lap before and after fuel correction (right). After correction the rising trend is tyre wear; the drops are the pit-stop waves.*

3. Computational approach

Workflow. One command, scripts/run_pipeline.py, runs eight stages in order: the numerical-methods study → data loading, cleaning and fuel correction → cross-validation (or, after the test, verification of the frozen selection) → the single test evaluation → strategy optimisation → sensitivity analysis → the credibility audit → performance benchmarks. Every numerical choice lives in config.yaml and is validated on loading; every stage writes CSV tables and figures to reports/, and a manifest stores the SHA-256 hash of every result so that a re-run can prove it reproduced them. Each choice is logged with its options, evidence and validation in DECISIONS.md.

3.1 Numerical methods implemented from scratch

Table 4. *From-scratch methods: how they work, where they are used and how each was verified.*

Method	How it works	Used for / settings	Verification
Euler, Heun, RK4	advance y with 1, 2 or 4 slope evaluations per step; global error O(h), O(h²), O(h⁴)	wear ODE in the model fit; RK4 with 1-lap steps	observed orders 1.00 / 2.00 / 4.00 (Fig. 2)
Dormand–Prince RK45	7-stage embedded pair; the 5th- minus 4th-order solution estimates the error; RMS norm scaled by atol + rtol·|y|; accept if ≤ 1; h_new = h·clip(0.9·err^(−1/5), 0.2, 10); last stage reused (FSAL)	accurate ODE reference; rtol 1e-3, atol 1e-6	25 f-evaluations vs SciPy's 50 at rtol 1e-3, same accuracy
Bisection, secant, Newton	halve a sign-changing bracket (linear, guaranteed); secant line through the last two points (order ≈ 1.6); tangent line using f′ (order 2)	wear parameter b (root of dRSS/db), dT/dL = 0, break-even values; stop when step AND residual small, or |step| ≤ 4·eps·|x|	agree with each other and with SciPy; b found in 21 / 5 / 3 iterations
Finite differences	forward/backward error O(h), central O(h²); round-off grows as eps/h, so an optimal h exists	dRSS/db, dT/dL, sensitivities; h = eps^(1/2), eps^(1/3), eps^(1/4) × max(1, |x|)	default h sits at the measured error minimum (Fig. 2)
Gaussian elimination	LU factorisation with partial pivoting (largest pivot swapped up to avoid dividing by small numbers); 1-norm condition estimate	normal equations, $(X^TX)^{-1}$ for standard errors; warn if cond > 1e12	matches numpy.linalg.solve to 1e-12
Linear least squares	solve $X^TX\beta = X^Ty$; cond(X^TX) = cond(X)², so inputs are scaled to [−1, 1]; SE from $\sigma^2(X^TX)^{-1}$	every model fit; QR as the alternative	matches lstsq; scaling cuts cond(X) from 1.6e10 to 94 (degree 6)
KNN regression	distance to every training point; average the k nearest (ties included), uniform or 1/distance weights	non-parametric degradation, k = 10–400	identical to scikit-learn kd-tree

3.2 Degradation models

Every model has the form lap time = offset(stint) + d(tyre age) with d(0) = 0. The per-stint offset removes car, driver and compound base pace; it is eliminated before fitting by subtracting stint means (Frisch–Waugh–Lovell), so only the shape of d is estimated. Three families were compared with a no-degradation baseline: a polynomial of degree 1–3 without constant; a wear ODE dd/dn = a + b·d, whose b is found by a 21-point grid scan and bisection on dRSS/db (RK4 integration, checked against the exact solution a·expm1(bn)/b on every fit); and KNN on tyre age. Parameter uncertainty comes from a bootstrap that resamples whole stints (200 resamples, seed 42), because laps within a stint are correlated. On the training data the wear ODE gives a = 0.0693 s/lap and b = +0.0010 per lap, with a 95% interval for b that contains 0: there is no evidence of an accelerating 'cliff'.

3.3 Race-time model and optimisation

For stints of n_i laps on compounds c_i, the race time is T = Σ_i [n_i·base(c_i) + Σ_{a=1..n_i} d_{c_i}(a)] + stops·pit loss − k_fuel·N(N − 1)/2 with N = 70 laps. A dry race must use two compounds, and no stint may exceed the largest tyre age observed with at least 5 laps (C3 40 laps, C4 35 laps), so the model is never extrapolated (D63). The strategy refits the selected model on 2023–2025 (D56) and uses an order-balanced C4−C3 pace delta of −0.080 s (D60). One-stop races are optimised by brute force (the reference) and by a continuous relaxation, where each stint's sum is replaced by a midpoint integral and the root of dT/dL is found with secant and Newton (central-difference derivatives) and then rounded and checked. Two-stop races are enumerated exhaustively.

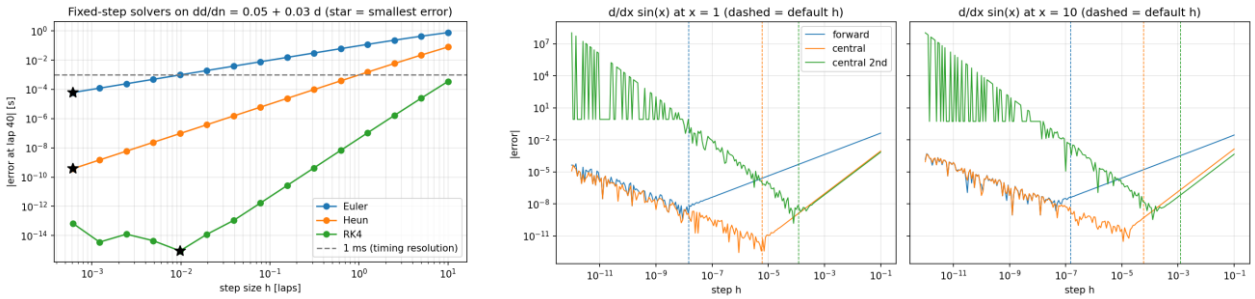


Figure 3. *Verification of the numerical methods. Left: error of Euler, Heun and RK4 against step size on the wear ODE (slopes 1, 2, 4). Right: finite-difference error against step h; the dashed lines (default steps) sit at the minimum between truncation and round-off error, i.e. the optimal step.*

4. Validation and verification

Code verification. 940 offline tests and doctests pass (pytest; HTML test report and 95% line coverage in the repository). Each from-scratch method is tested against an analytic solution or a trusted library, including its order of convergence. Tests use small synthetic tables and never download data.

Model selection: leave-one-race-out cross-validation on 2022–2024 (3 folds). Each held-out stint's base pace is set by its first 3 laps, which are not scored; all other laps are predicted. RMSE is the primary metric. A candidate is accepted if it beats the no-degradation baseline in every fold, its mean bias is within ±0.10 s/lap, and its curve is non-negative and does not fall by more than 0.05 s between laps. The simplest accepted family winner within one standard error of the best RMSE is selected. The selection was frozen in model_selection.json (11:40:30 UTC) before the single test run (11:40:44 UTC); a code 'firewall' refuses to repeat the test or overwrite the selection.

Table 5. *Cross-validation results (mean ± standard error over folds).*

Model	C3 CV RMSE [s]	C3 bias [s/lap]	C4 CV RMSE [s]	C4 bias [s/lap]	Accepted?
Baseline (no degradation)	1.168	−0.866	1.132	−0.634	no (reference)
Polynomial degree 1	0.633 ± 0.091	+0.040	0.851 ± 0.151	+0.084	yes – **selected**
Polynomial degree 2	0.633	+0.035	0.849	+0.134	C3 yes, C4 bias fails
Wear ODE	0.633	+0.035	0.849	+0.129	C3 yes, C4 bias fails
KNN k = 100, uniform	0.639	−0.007	0.851	−0.012	yes

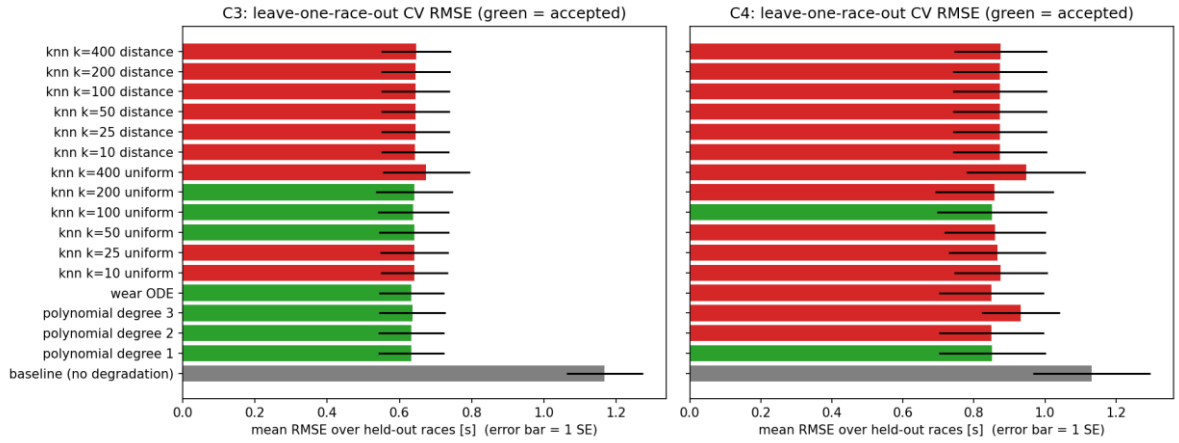


Figure 4. *Cross-validated RMSE of all 17 candidates per compound. The flexible models do not beat a straight line, so the one-standard-error rule selects degree 1 for both compounds.*

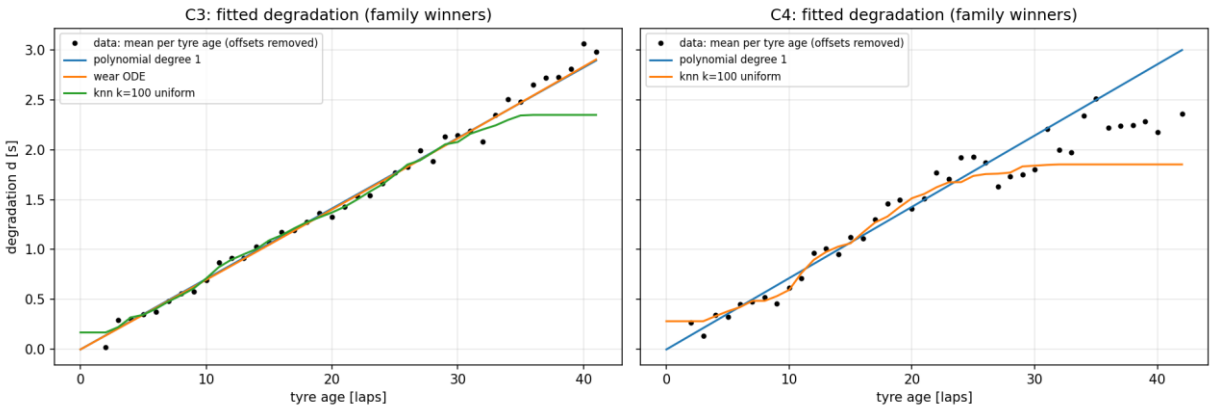


Figure 5. *Tyre degradation fitted to the training races by the family winners (C3 left, C4 right; points = mean per tyre age after removing stint offsets): the straight line, the wear ODE and KNN (k = 100) coincide up to about 30 laps; beyond that KNN flattens because few laps remain to average.*

4.1 Test on the withheld 2025 race and its diagnosis

On 2025 the selected lines had RMSE 0.942 s (C3) and 1.042 s (C4), better than the baseline (1.311 and 1.082 s), but a positive bias of +0.276 and +0.536 s/lap: they predicted more degradation than occurred, and the 0.10 s/lap criterion was **not** met. A post-test diagnosis that changes nothing (D89) splits the bias exactly into a wear-rate part and an anchor-lap part (Table 6). The 2025 wear rate was 30–36% lower than in 2022–24 (the 2025 C3 was softened and the median track temperature was 13–17 °C lower). For C4, the first laps of the 2025 stints were also about 0.2 s slow, which shifts every anchored prediction. Extrapolation beyond the training ages, the fuel correction (leftover effect −0.0027 s/lap, CI includes 0) and any single team (14 of 20 team groups biased positive) were ruled out.

Table 6. *Exact decomposition of the 2025 test bias [s/lap].*

Compound	Wear rate train → 2025 [s/lap]	Observed bias	Wear-rate part	Anchor-lap part	Bias within training ages
C3	0.0705 → 0.0451	+0.276	+0.416	−0.140	+0.268
C4	0.0713 → 0.0496	+0.536	+0.297	+0.239	+0.536

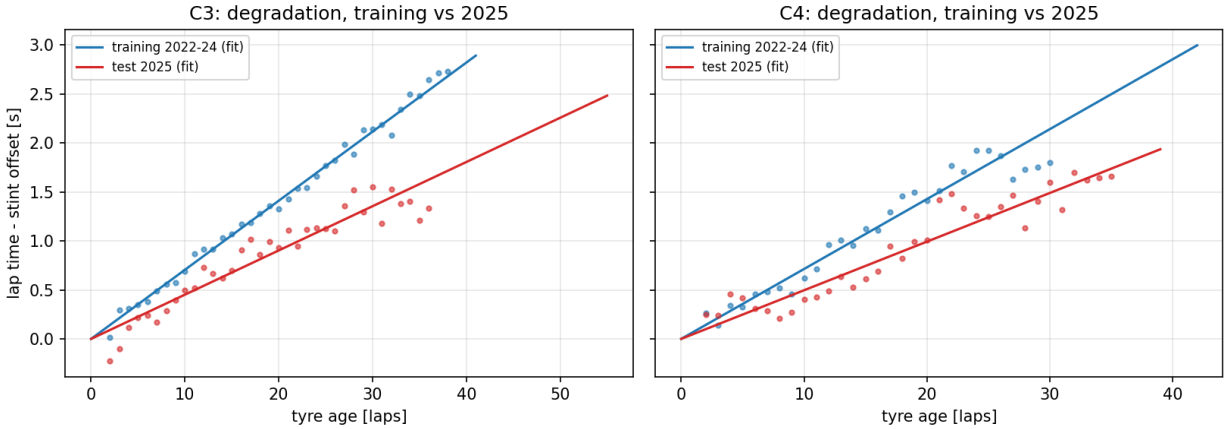


Figure 6. *2025 diagnosis: observed degradation (points, mean per tyre age after removing stint offsets) and fitted lines, training 2022–24 versus the 2025 test race. The 2025 slopes are 30–36% lower.*

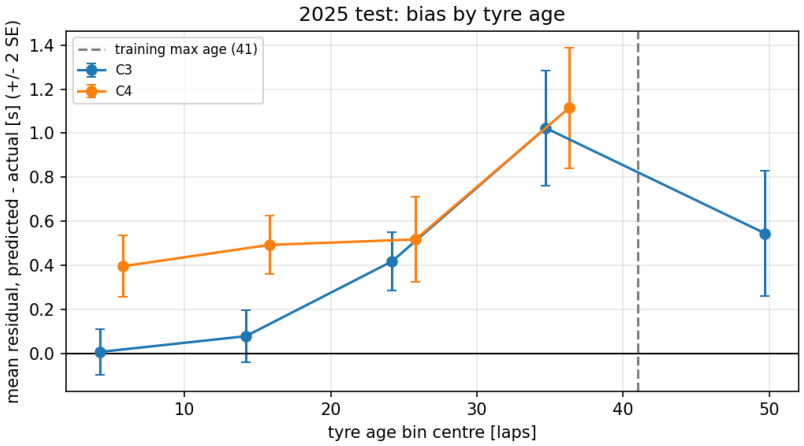


Figure 7. *Mean 2025 test residual by tyre-age bin (±2 SE). C3 grows with age (slope too steep); C4 is already +0.4 s at young ages (anchor effect).*

4.2 Leakage, constraint and reproducibility audit

An audit stage (D90) re-checks the real results on every pipeline run and stops the run if a check fails. All 14 checks pass: training and test races are disjoint; the selection was fitted without 2025 and frozen and committed before the test (git ancestry); each race is cleaned and fuel-corrected independently; anchor laps are never scored (1,730 scored, 261 anchor, overlap 0); the training races alone imply k_fuel = 0.0452 s/lap. For the strategy, every ranked strategy obeys the stint limits and the two-compound rule, its time matches an independent stint-by-stint sum within 1e-6 s, and an exhaustive search of every stint split confirms the best one-stop and two-stop and shows that the best three-stop race (5935.14 s) is slower. Two consecutive runs reproduce all 52 result files byte for byte, and so does a fresh clone from GitHub that downloads the data from scratch.

5. Alternative solutions and comparison

Table 7. *Alternatives considered and the evidence behind each choice.*

Decision	Alternatives compared	Evidence	Choice
Degradation model	baseline, polynomial 1–3, wear ODE, KNN (12 settings)	CV RMSE within 1 SE; bias; physical checks (Table 3)	polynomial degree 1: simplest, accepted
ODE solution in the fit	Euler, Heun, RK4, RK45, exact	cost to reach 1e-8 s: RK4 256, RK45 103 f-evals; exact 0.009 ms	RK4 (1-lap steps), verified against exact each fit
Least-squares solver	normal equations (mine), QR, SVD	degree 6 unscaled: 3.6e-9 / 5.1e-13 / 1.7e-10 rel. error; scaled: all ≤ 2.3e-13	normal equations + input scaling
Root finder for b	bisection, secant, Newton	21 / 5 / 3 iterations, identical a and b	bisection (guaranteed bracket)

Decision	Alternatives compared	Evidence	Choice
One-stop optimiser	brute force, secant, Newton	6 / 9 / 18 evaluations with limits; 69 / 12 / 21 without	brute force reference; secant 5.9× faster unconstrained
KNN search	brute force (mine) vs kd-tree	same predictions; time ∝ n^1.07 vs n^0.23; 270 MB vs 1.4 MB at n = 16,000	brute force (exact, simple; data small)
Data source	F1 TV + FastF1 pre-release, free data	free sources reproducible by anyone	FastF1 mirror + OpenF1

6. Strategy results

With the 2023–2025 refit (C3 0.0597 s/lap, C4 0.0551 s/lap), the best strategy for the 70-lap race is **two stops, C3 → C4 → C4 with stops after laps 21 and 45: 5926.50 s**, 4.33 s faster than the best one-stop (pit lap 35, 5930.83 s). For the one-stop, the relaxed optimum dT/dL = 0 lies at lap 32.89, but the C4 stint limit (35 laps) moves the feasible optimum to lap 35: the data-coverage rule, not the physics, sets this pit lap (C5, C9). Brute force, secant and Newton agree. Several two-stop orders tie because the race-time curve is flat near the optimum (C6).

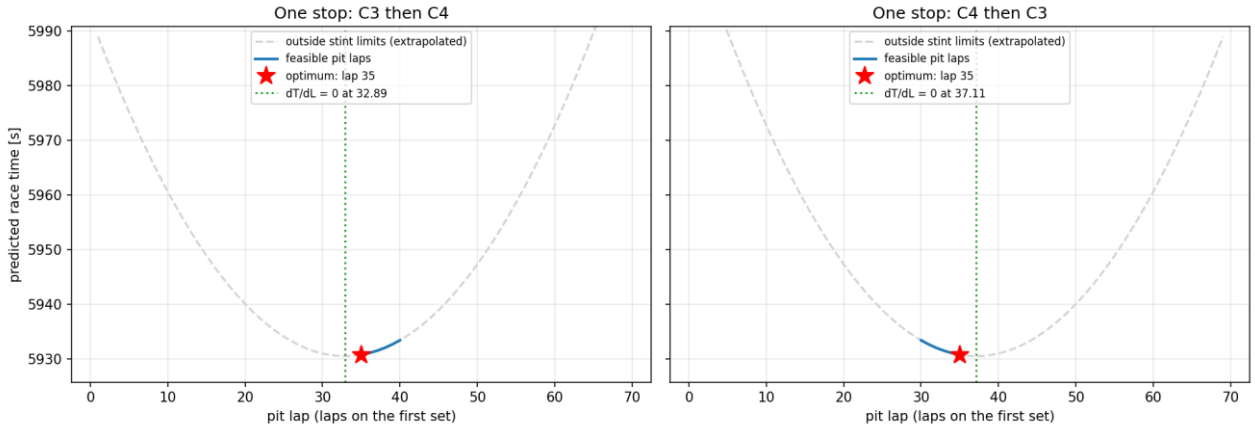


Figure 8. *Predicted race time against pit lap for both one-stop orders: grey = outside the stint limits (extrapolated), star = optimum (lap 35), green = continuous root of dT/dL (lap 32.89).*

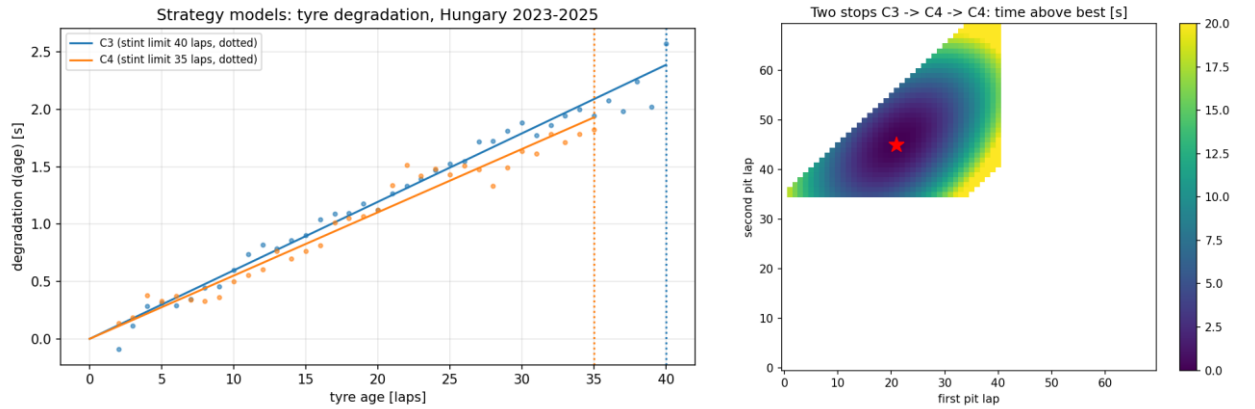


Figure 9. *Left: the degradation models used by the strategy, with the observed data and the stint limits (dotted). Right: two-stop race C3 → C4 → C4, time above the best for every pair of pit laps; the star marks laps 21 and 45, white cells break a stint limit.*

7. Sensitivity analysis

Each uncertain input was varied alone over a physically motivated range (D67), and local coefficients were computed with central differences at a 1% step, checked at 0.1% and 10% (D68). Exact checks confirm the method: d(one-stop time)/d(C3 rate) = 630 = 35·36/2 and d(margin)/d(pit loss) = −1. Bisection gives the break-even values where one and two stops are equal: **pit loss 25.22 s** (outside its IQR, so it cannot flip the decision) and **k_fuel 0.0333 s/lap** (inside its plausible range, so it can). Year-to-year wear also flips it: in a 2025-like low-wear year the one-stop wins by 1.27 s. Within a stint, lap number and tyre age rise together, so the fitted wear rate changes one-for-one with k_fuel (d rate/d k_fuel = 1): fuel and wear cannot be separated, and the k_fuel derivatives depend on the step size (1149 / 1158 / 745 s per unit at 0.1 / 1 / 10%), so they are reported as indicative only (C8).

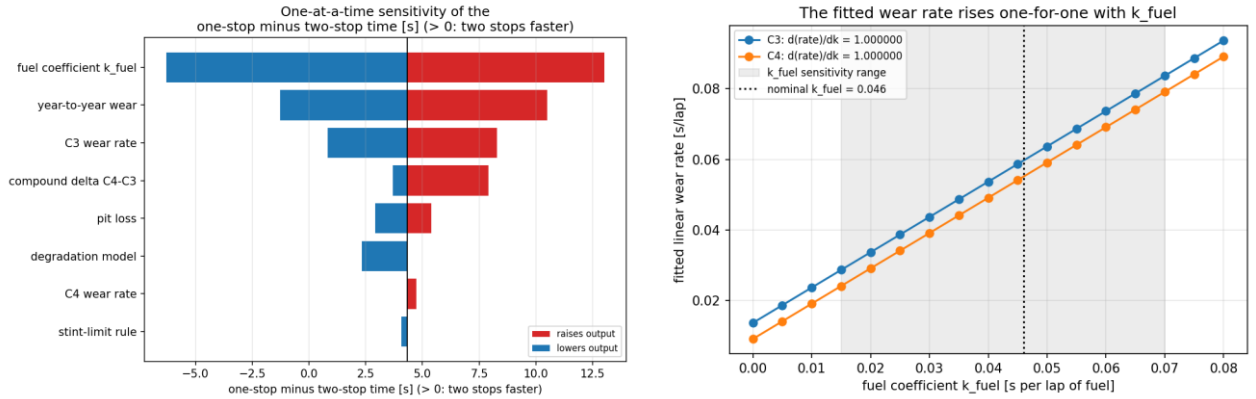


Figure 10. *Left: tornado plot of the two-stop advantage (one-stop minus two-stop time, nominal +4.33 s); only the fuel coefficient and year-to-year wear can make it negative. Right: the fitted wear rate rises exactly one-for-one with the fuel coefficient, so fuel and wear cannot be separated.*

8. Performance and optimisation

Timings use repeated runs whose repeat length is calibrated to at least 10 ms (7 repeats, minimum reported), memory uses tracemalloc, and FLOPs are counted per step: Euler f + 3, Heun 2f + 7, RK4 4f + 17 and one RK45 attempt 105 FLOPs for the wear ODE (f = 2). On the ODE benchmark RK4 reaches 1 ms-level accuracy with 16 f-evaluations, Euler needs 4096; RK45 reaches 1e-8 with 103. RK4 runs at only 1.35 MFLOP/s (4.6 µs per evaluation), so run time is dominated by Python overhead, not arithmetic, which is why the closed-form exact solution (0.009 ms) is used as the check. cProfile showed that the cross-validation's prediction step took 67% of the compute time because it looped over every held-out stint. Vectorising it (one model call and numpy.bincount for the anchors) cut the pipeline from 2.16 ± 0.04 s to 1.75 ± 0.02 s (1.23×) with identical results (max difference 5.6e-16, tolerance 1e-12 s, enforced by a test). The profiler had suggested a larger gain because it adds overhead to every small call (C14).

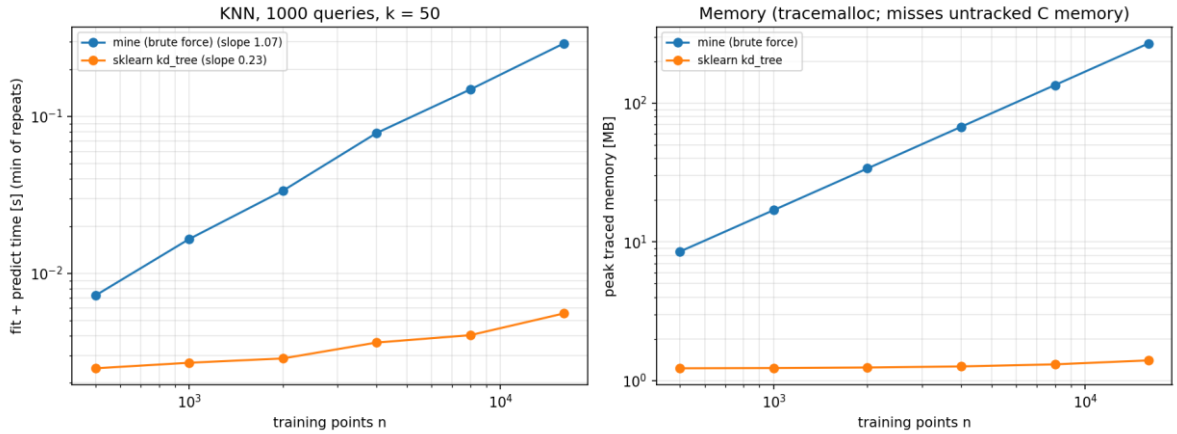


Figure 11. *KNN scaling: run time and memory against training-set size; my brute-force search grows linearly (exponent 1.07), scikit-learn's kd-tree sub-linearly (0.23), with identical predictions.*

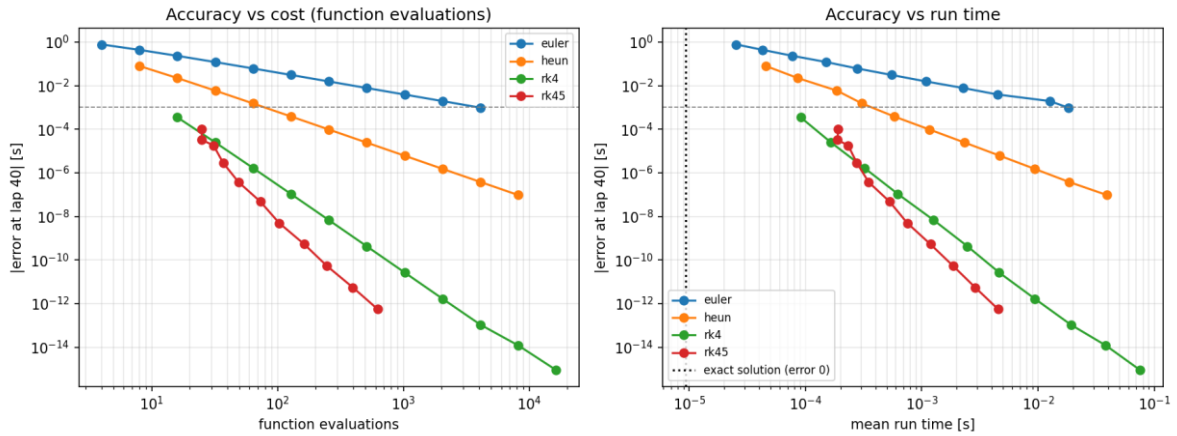


Figure 12. *Accuracy against cost for the ODE solvers: error versus function evaluations (left) and versus run time (right).*

9. Challenges and how they were resolved

Every problem was logged when it happened, with how it was detected, its root cause and its resolution (CHALLENGES.md; Phase 1–5 problems are recorded in DECISIONS.md). Table 8 lists all of them; the numbers in brackets are the log entries.

Table 8. *All challenges met during the project, with root cause and resolution.*

#	Challenge	Root cause	Resolution / lesson
1	F1 timing archive locked (OI-1, D14)	from mid-October 2026 the archive requires an F1 TV login; FastF1's free mirror has 2021–22 only	FastF1 2022 + OpenF1 2023–25 in one unified table; every response cached
2	OpenF1 refused requests (D15)	HTTP 429 when too fast; HTTP 401 during live F1 sessions	0.4 s delay, retries with back-off, clear error for live sessions, disk cache
3	Scope changed three times (D4, D6, D52)	Bahrain had no Red Bull HARD laps (2022) and no MEDIUM laps (2023–24); one team gave too few stints	data surveys; select by Pirelli C-number; Hungary C3 (+ C4), all teams with per-stint offsets
4	Fuzzy event matching (D7)	FastF1 silently returned the Australian GP for an invented event name	the loader compares the resolved name and raises RaceNotFoundError
5	YAML read 1.0e10 as text (D38)	YAML needs a sign in the exponent	write 1.0e+10; config.py rejects text values
6	Stopping rule never stopped (D30)	requiring step AND residual small fails when the floats nearest the root still give $\|f\| > f_{tol}$	round-off safeguard $\|step\| \le 4 \cdot eps \cdot \|x\|$; flat and steep cases tested
7	Compound pace hidden (C1)	per-stint offsets absorb compound pace as well as car pace	estimate the compound delta separately
8	Delta said C4 slower than C3 (C2)	35 of 43 driver-races ran C4 first: stint order confounded with compound	order-balanced delta −0.080 s, range −0.32 to +0.15 s in the sensitivity
9	Doctest placeholders (C3)	comment-only examples are not valid doctests	real examples; doctests run with every test suite
10	Test fixture wrongly constrained (C4)	the test fixture's stint limit, not the code, rejected the stints	fixture fixed; confirms the limit is enforced everywhere
11	One-stop optimum pinned (C5, C9)	the C4 stint limit (35 laps) is below the relaxed optimum	report the active constraint; pit lap set by data coverage
12	Flat curve, tied strategies (C6)	race time changes little near the optimum	ties reported as ties; margin compared with input uncertainty
13	No feasible one-stop (C7)	with a ≥ 10-lap rule the limits sum to 67 < 70 laps	infeasible scenarios handled explicitly; a genuine finding
14	Step-dependent derivatives (C8)	outputs pass through medians and an integer optimum (not smooth)	step-size check at 0.1/1/10 %; k_fuel derivatives marked indicative
15	Figure title cut off; bad edit (C10)	long title; a shell edit inserted a literal newline	title shortened; no escape-interpreting edits
16	Hand-counted FLOPs wrong (C11)	an expected doctest value was typed, not computed (104 vs 105)	only computed values are used as expected results
17	Noisy timings (C12, C13)	20–100 µs calls are at the scale of OS jitter	repeats calibrated to ≥ 10 ms; report the minimum; conclude only from ≥ 2× differences
18	Profiler overstated gain (C14)	cProfile adds overhead to every small call	speed-up confirmed with plain timings (1.23×)
19	Disk-cache effect on I/O timing (C15)	the operating system caches files after the first read	warm time reported; data loading excluded from the optimisation
20	Frozen selection overwritten (C16)	a verification re-run of the CV stage rewrote model_selection.json after the test	restored from git; guard refuses a CV re-run after the test unless forced
21	Local paths leaked (C17, C19, C20)	profiler, pdoc and pytest-html wrote absolute paths with the user name	relative paths, all artefacts scanned, git history rewritten (D92)
22	Last-digit differences (C18)	vectorised sums in a different order change round-off (≤ 6.1e-16)	results regenerated; 'same result' always stated with a tolerance
23	Cleaning report showed 0 laps (C21)	the 'input' row has removed = 0 by definition	laps loaded, removed per rule and kept, which now add up
24	126 incomplete docstrings (C22)	rule not checked automatically while coding	review script → 0 gaps; 18 functions given input checks with 49 tests
25	Stale docstring (C23)	a Phase 2 note contradicted the later strategy decision D58	corrected; audit check B9 explains why it is not leakage
26	Reproducibility check skipped in a clone (C24)	git on Windows checks text files out with CRLF, changing the hashes	text hashes ignore line endings; fresh clone reproduces all 52 files
27	2025 test bias criterion failed (D89)	2025 wear 30–36 % lower; slow anchor laps for C4	exact bias decomposition; 2023–25 refit; year scenarios in the sensitivity

10. Limitations and lessons learned

- **One circuit and two compounds.** The parameters do not transfer to other tracks; four races give only three CV folds and one test race.

- **Year-to-year change dominates.** The test bias and the negative test R^2 (−0.04 C3, −0.47 C4) come from a season with 30–36% lower wear; cross-validation within past seasons cannot detect this, and the strategy decision can flip with the year.
- **Fuel and wear are confounded.** One k_fuel for all races, which also contains track evolution; the two-stop decision flips below 0.0333 s/lap.
- **Not modelled:** track temperature, weather, car set-up, driver, traffic and dirty air, tyre warm-up, safety cars, tyre-set availability and used tyres.
- **Lessons:** select data by the physical quantity, not the label; protect the test set in code; decompose a bias before explaining it; report active constraints with the optimum; check numerical derivatives for step-size dependence; never trust an expected value that was not computed; and check every generated file before publishing it.

11. AI-use reflection

Table 9. *AI-use reflection (also in the repository README).*

Item	Statement
Tool	Claude Code (Anthropic), VS Code extension, model Claude Opus 5.5, used for the whole project.
Contribution	The AI wrote essentially all of the code, the 940 tests and the first drafts of the documentation and this report, and ran every analysis. My contribution was the project specification and working rules, every Phase 1–5 decision, the questions that changed the analysis, and the review of each phase before it was committed.
Why and how	Writing every method from scratch with tests is a large amount of code. To keep control of the engineering, I required the AI to list every decision with options, a recommendation, evidence and a validation plan, and to stop until I approved or changed it (Phases 1–5). For Phases 6–9, I let it decide, with every such decision marked ‘to be reviewed by me’ in DECISIONS.md.
My decisions	Data surveys of every track and compound before choosing Hungary/C3 (later + C4); all teams instead of Red Bull; free data only; a step-size study for the optimal finite-difference step and comparisons of alternative methods; the 1e12 conditioning threshold; the Phase 2 cleaning options; approving the root-finder safeguard after my first choice failed; questions on track temperature and missing wear factors; how to respond to the failed test criterion; public repository with history rewrite.
Verification	Benchmarks against SciPy, scikit-learn and analytic solutions with orders of convergence; withheld-race test behind a code firewall; leakage and constraint audit with an independent exhaustive search; byte-for-byte reproduction from a fresh GitHub clone; I read each phase summary of assumptions and uncertainties before approving, and asked for explanations (e.g. the IQR, the purpose of a validation test).
AI errors	Unverified doctest values (C3, C11); a re-run that overwrote the frozen selection (C16); local paths leaked three times (C17, C19, C20); 126 incomplete docstrings against the project’s own rule (C22); a stale docstring (C23); a first stopping rule that could fail to stop (D30).
Effect on learning	The decision protocol made the AI explain options that I had to choose between, which is how I learned why central differences beat forward differences, why the normal equations lose accuracy with conditioning, and why within-season cross-validation cannot detect a season change. Its mistakes taught me that plausible output is not verified output.

12. Conclusion

Tyre degradation at the Hungaroring is well described by a straight line in tyre age once fuel, pit laps and non-green laps are removed and each stint has its own base pace: cross-validated RMSE 0.63 s (C3) and 0.85 s (C4), with no gain from polynomial, ODE or KNN models. On the withheld 2025 race the model still beat the baseline but over-predicted wear, which the audit traces to a 30–36% lower 2025 wear rate (and, for C4, slow first laps). The verified optimum for a typical car is a two-stop C3 → C4 → C4 race, 4.33 s faster than one stop, but this advantage reverses for a fuel coefficient below 0.0333 s/lap or in a low-wear season. The value of the work lies as much in the verified numerical toolkit, the honest test and the audit as in the strategy itself.

References

Dormand, J. R. and Prince, P. J. (1980). A family of embedded Runge–Kutta formulae. Journal of Computational and Applied Mathematics, 6(1), 19–26.

Efron, B. and Tibshirani, R. J. (1993). An Introduction to the Bootstrap. Chapman & Hall.

Sen, P. K. (1968). Estimates of the regression coefficient based on Kendall’s tau. Journal of the American Statistical Association, 63, 1379–1389.

FastF1 documentation. https://docs.fastf1.dev/ ; OpenF1 API. https://openf1.org/ (accessed October 2026).

Pirelli press: 2022 Hungarian Grand Prix preview, https://press.pirelli.com/2022-hungarian-grand-prix---preview/ ; Tyre news and nominations for Hungary and Belgium (2023), https://press.pirelli.com/tyre-news-and-nominations-for-hungary-and-belgium/

Formula1.com: What tyres will the teams and drivers have for the 2024 / 2025 Hungarian Grand Prix (URLs in reference/pirelli_allocation.csv).

SciPy, scikit-learn and NumPy documentation (used as benchmarks only).

Monash University. MMA3001 Numerical Methods and Machine Learning — Project Brief (2026).

Appendix A. Remaining figures

All 24 figures are regenerated by one run of scripts/run_pipeline.py; the figures not shown in the main text follow.

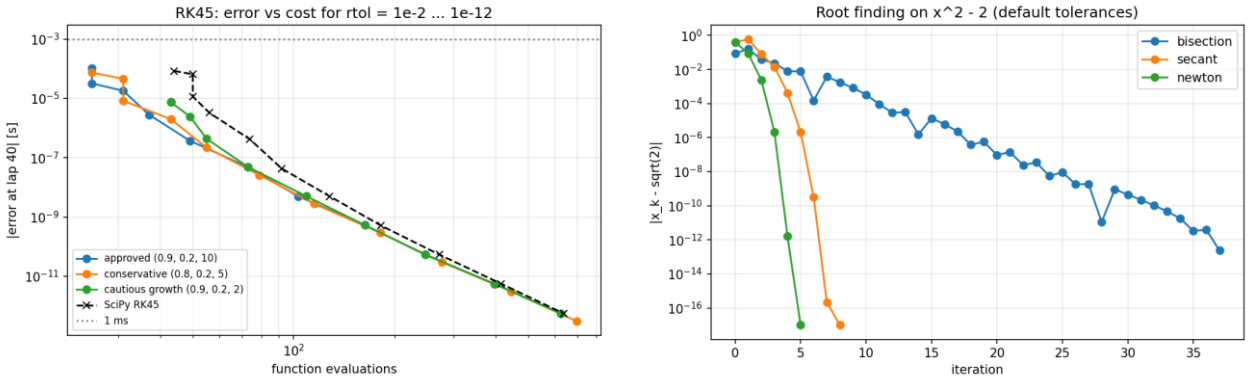


Figure 13. *Left: my RK45 (three step-size controllers) against SciPy's RK45, error against function evaluations for rtol 1e-2 to 1e-12. Right: bisection, secant and Newton on $x^2 - 2$ with the default tolerances, error against iteration (linear, super-linear and quadratic convergence).*

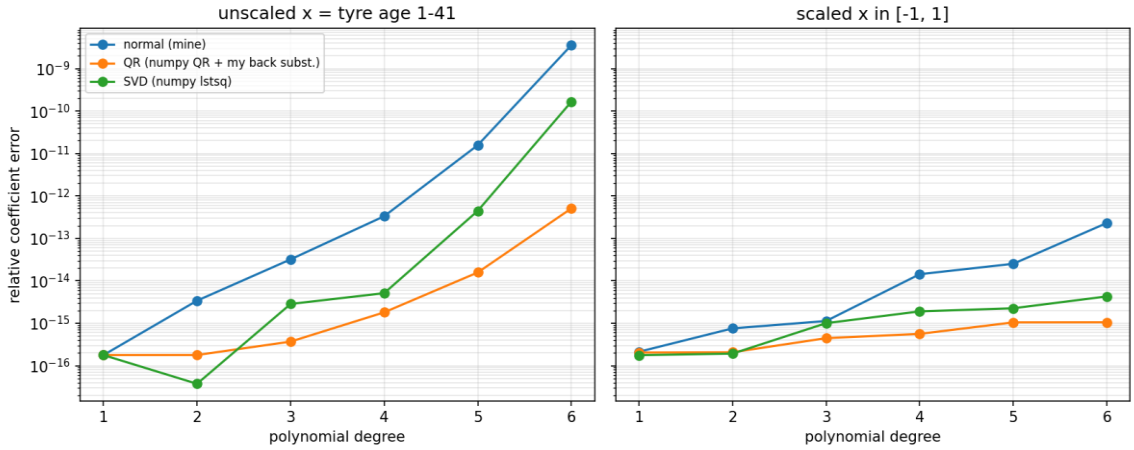


Figure 14. *Least squares: relative coefficient error against polynomial degree for the normal equations, QR and SVD, with and without input scaling.*

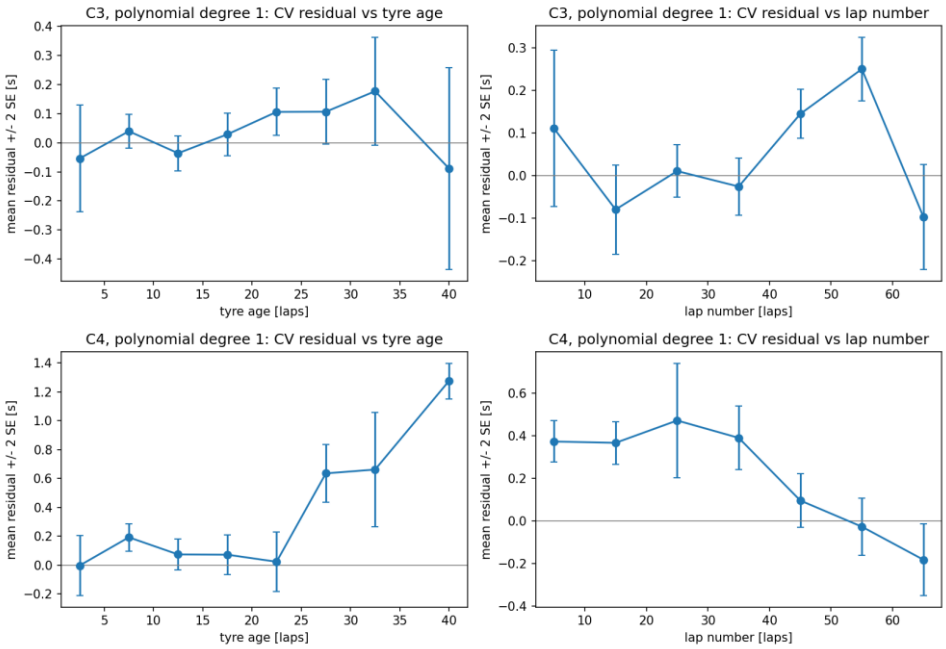


Figure 15. *Cross-validation residuals by tyre age and lap number (leftover fuel effect check).*

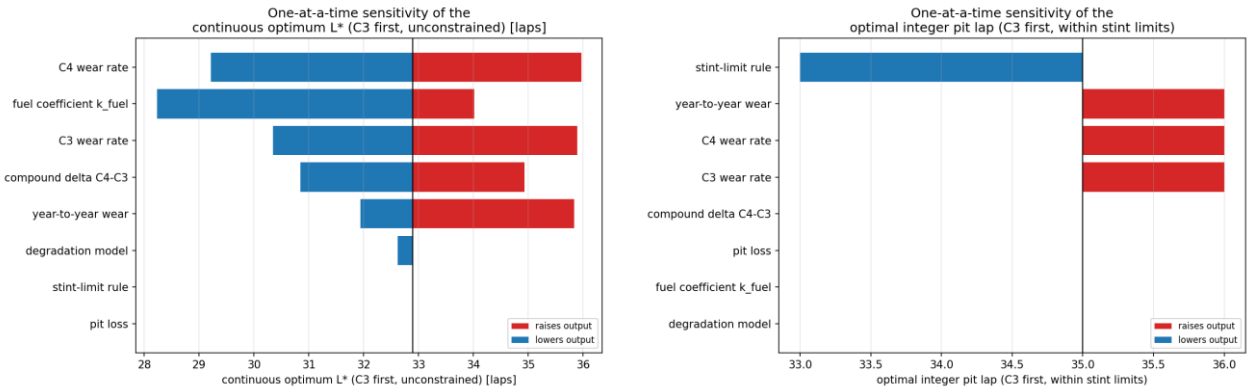


Figure 16. *Tornado plots for the continuous optimum L* (left) and the integer pit lap (right), which the stint limit pins at 35–36.*

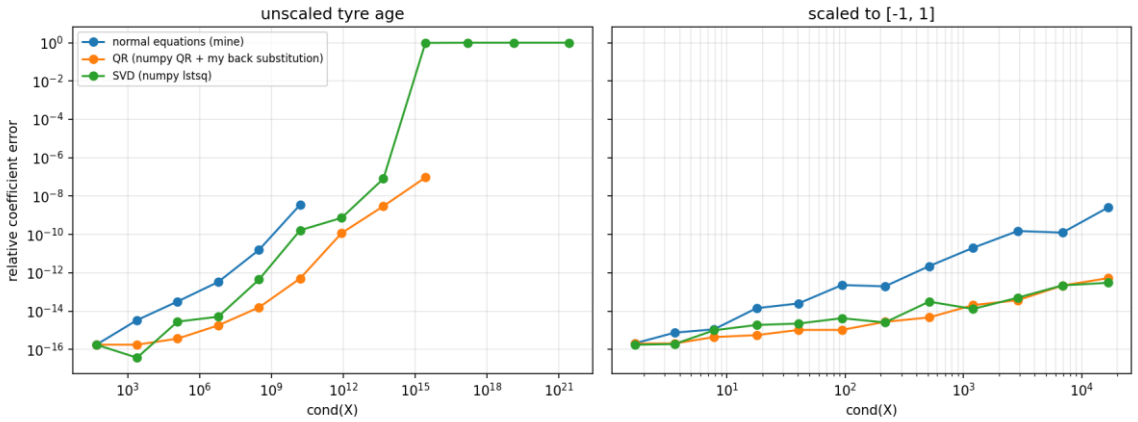


Figure 17. *Least squares, degrees 1–12 on tyre ages 1–41 (Phase 8): relative coefficient error against cond(X). Unscaled (left), my normal equations refuse as singular from degree 7 and QR from degree 10, while SVD (lstsq) never refuses but is silently 98–100% wrong from degree 9. Scaled to [−1, 1] (right), every method stays accurate.*